RDB-YOLO: A Lightweight and High-Precision Steel Surface Defects Detection Method

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***Abstract*—Most deep learning-based target detection algorithms have low speed and accuracy for detecting defects on steel surfaces, which fails to meet the requirements of industrial sites. To tackle this important issue, we propose a lightweight network structure based on YOLOv5n. Firstly, Receptive Field Enhancement (RFE) is introduced into the backbone network for compensation for loss of accuracy caused by light changes and environmental noise interference. Secondly, Deformable Convolution network (DCN) is utilized to improve performance of the network to fit defects of different shapes. Thirdly, Bi-level Routing Attention mechanism (BRA) is embedded into the neck to improve detection performance in small targets. Experimental data from the NEU-DET dataset indicates that the Mean Average Precision (mAP) of the improved algorithm is 82.1%, which is 5.1% higher than that of the benchmark model, and the detection rate is 60.4 frame/s. Compared with other mainstream object detection algorithms, our algorithm has higher accuracy, lighter weight parameters and can achieve efficient steel surface defect detection under the premise of maintaining a certain detection speed.**

Keywords—Surface defect detection, YOLOv5n, Deformable Convolution, RFE, BRA

I. INTRODUCTION

Being among the most widely used materials in the manufacturing sector, metal remains highly prevalent, steel has been used across numerous applications, such as aerospace, machinery manufacturing, petrochemical and other fields, along with the development of the country's industrialization. However, due to its thin characteristics, steel strip is highly susceptible to certain defects, including cracks, scratches, patches, etc., which are mostly manifested as surface defects during the process of rapid rolling [1]. Cracks, inclusions, scratches and other common steel surface defects will have a significant impact on the fatigue resistance and corrosion resistance and even increase the incidence of safety accidents, therefore, identifying surface defects on steel holds considerable importance.

Compared with traditional steel surface defect detection, deep learning methods can extract more abstract and adaptive features from the data, resulting in better performance in various situations. Hao et al. [2]proposed a deformable convolution-based steel defect detection algorithm with a mAP of 80.6%, which demonstrated the applicability of deformable convolution in steel surface defect detection. Although the two-stage target detection algorithm can have better performance on accuracy, the detection speed is slow due to the need to extract features for each candidate frame. This limitation poses a challenge in meeting the real-time requirements in industry. The single-stage algorithm directly predicts the classification results and positional coordinates of the target through regression methods, which is characterized by high accuracy and speed. Chen et al. [3] proposed a network structure based on YOLOv3, which introduced a dense convolutional block in the network to effectively improve the precision in detecting small-scale defects and unclear features. Liu et al. [4] proposed a single-stage network built on MobileNet and depth-separable convolution to improve performance on accuracy and speed. Zhong et al. [5] embedded Convolutional Block Attention Module (CBAM) in YOLOv5 and achieved a mAP of 91.0% on the self-constructed dataset. Cheng et al. [6] introduced a RetinaNet model incorporating differential channel attention alongside adaptive spatial feature fusion, which can effectively suppress complex background interference. Yu et al. [7] introduced an efficient network with multi-scale awareness to address issues related to information loss in detecting tiny defects, as well as mismatches between the detection head's receptive field and the target scale. Zhou et al. [8] introduced a detection network to alleviate the accuracy degradation caused by object shape. Although Significant progress has been made in the field of steel defect detection through deep learning, there is still considerable potential for improvement in real industrial environments due to the noise generated by industrial cameras, ambient light sources, and other uncontrollable factors[9][10] resulting in reduced detection accuracy, coupled with the fact that steel surface defects exhibit the complex shapes and variable sizes, which makes detection more difficult.

In this paper, an RDB-YOLO algorithm, leveraging DCN and an attention mechanism, is developed to expand the receptive field, suppresses the environmental noise, and improves the detection performance of irregular defects and small defects on steel surface. For irregular defects, deformable convolution [11] is used instead of traditional convolution, which improves the network's performance to detect defects across various scales. A multi-scale awareness module is employed, which is utilized to increase the receptive field, thus suppressing noise and improving the detection accuracy. In addition, BRA mechanism is used in the neck to enhance the accuracy of detecting small defects. The primary contributions of this paper are outlined as follows.

(1) First, considering the diversity of defects on steel surfaces, including variations in morphology and size, adaptive deformation convolution is employed in the feature fusion module of the model to enhance the model's adaptation to these defects by learning the positional offsets within the convolution.

(2) Second, we develop a novel module with multi-scale awareness module to expand the receptive field of the feature map by merging dilated convolutions with different expansion speeds,

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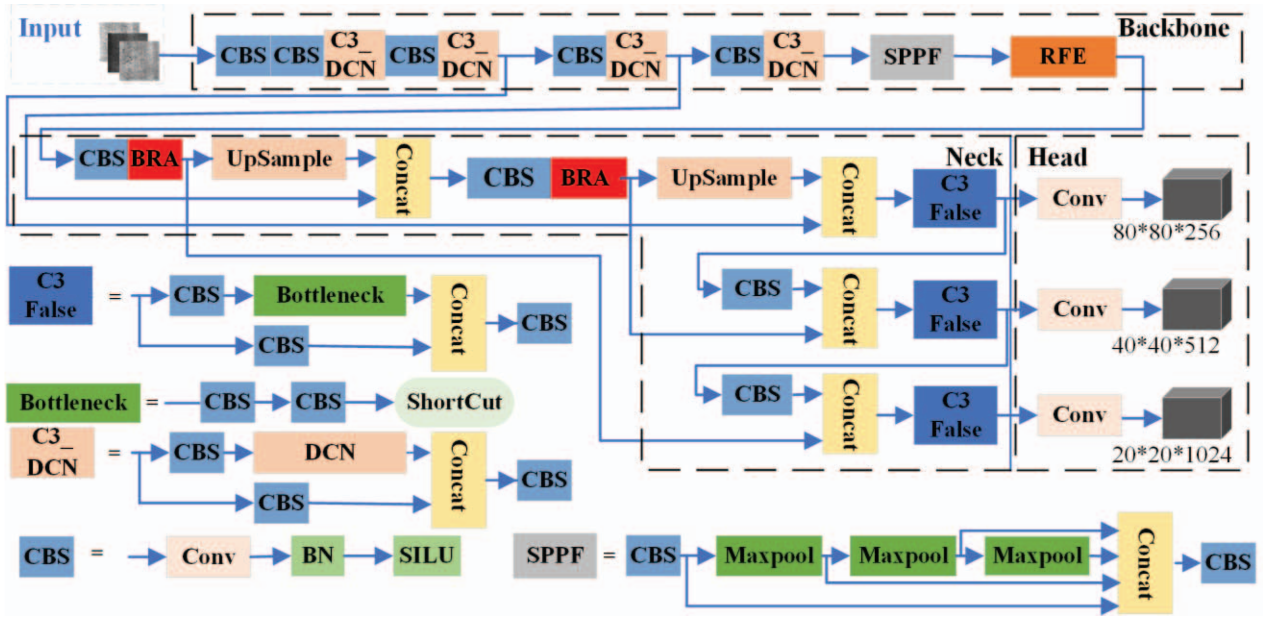

Fig. 1. Structure of the RDB-YOLO.

which mitigates the noise interference and improves the robustness and accuracy of the mode.

(3) Finally, considering small defects in the image, BRA mechanism is integrated into the model, using sparse sampling rather than downsampling, which can preserve fine-grained information and improve the precision of detection of small defects.

The rest of the paper is organized in the following manner: Section 2 proposes a new defect detection network, RDB-YOLO. Section 3 briefly describes the steel surface defect dataset and conducts various experiments focused on performance evaluation and relevance of the RDB-Net network. Section 4 summarizes the main work and outlines future work.

II. Methods

The YOLOv5 is a classic and popular single-stage detector with high detection accuracy and high speed, making it widely used in identification of targets. The YOLOv5 algorithm is made up of five versions. The general framework of the algorithms remains the same in all five versions, excluding the model's depth and width. The YOLOv5 network structure is made up of three parts: the backbone, the neck and the detection head. The backbone network handles the extraction of image features, the neck is responsible for the fusion of features from different scales, and the head is tasked with predicting the target category and location. Considering the strict requirements for defect detection speed in industrial environments, YOLOv5n has been chosen as the baseline model for steel defect detection, as it is recognized for its fast detection speed and relatively high detection performance on accuracy.

This paper proposes RDB-YOLO model, and the configuration of the network is displayed in Fig. 1. Firstly, deformable convolution is used in the backbone of the network instead of traditional convolution to extract features of complex defects and enhance the flexibility of the network. Secondly， the RFE module is designed between the backbone and the neck for enhancing the receptive field and thus suppressing noise. Finally, BRA is added to the neck, which increases the fine-grained information and improves the detection performance of small defects since BRA is based on sparse sampling instead of downsampling.

A. *Deformable Convolution network*

Traditional convolution is applied for extracting features in steel surface defect detection, and its sampling location can be limited by the shape of the rectangle. It is difficult for the receptive field to capture the geometry of irregular objects, which may cause the feature map to lose key information through the extraction procedure, weakening the competence of the network model to detect irregular defects. To effectively solve this problem, this paper enhances the model's ability to adapt to irregular defects by introducing deformable convolution network.

The calculation of DCN is as follows:

$$y(p_0) = \sum_{k=1}^{K} w_k \cdot x\left(p_0 + p_k + \Delta p_k\right) \cdot \Delta m_k \tag{1}$$

where $y(p_0)$ is the output of deformable convolution, p_0 and p_k are the locations of the current point x and its neighbors, Δp_k represents the learnable offset of the deformable convolution, w_k r denotes the weight associated with the p_k point, and Δm_k represents the modulation scalar of the weights for the k-th sampling location. The modulation factor Δm_k falls within the rang [0, 1], and Δp_k is real number with unconstrained range.

Specifically, by leveraging the learned offsets, the deformable convolution kernels can adjust to scale, rotation, and other intricate geometric transformations of the defects. As shown in Fig. 2, the green arrows show the offsets learned

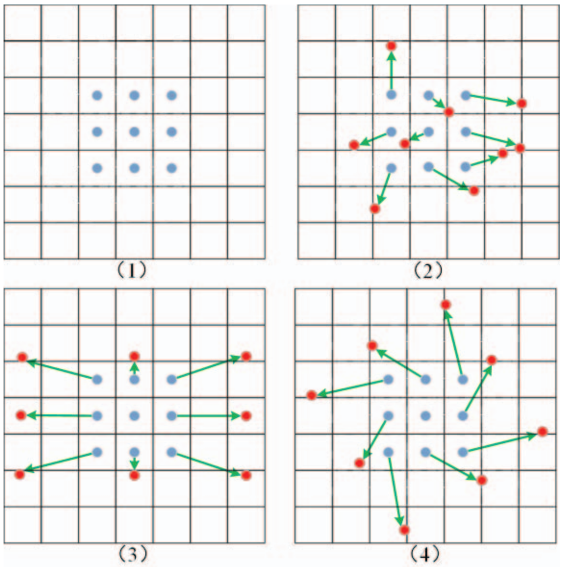

Fig. 2. Standard convolution and Deformable Convolution v2, (1) is the sampling grid of the standard convolution, (2), (3) and (4) represent the Deformable Convolution with different sampling methods.

through deformable convolution. Red points represent the learned deformable sampling locations and bule points represent the standard convolution sampling locations.

The specific structure of deformable convolution is shown in Fig. 3. The offsets are determined from the previous feature maps through the use of additional convolutional layers in parallel, where the size of the output offsets is the same as the size of the input feature tensor.

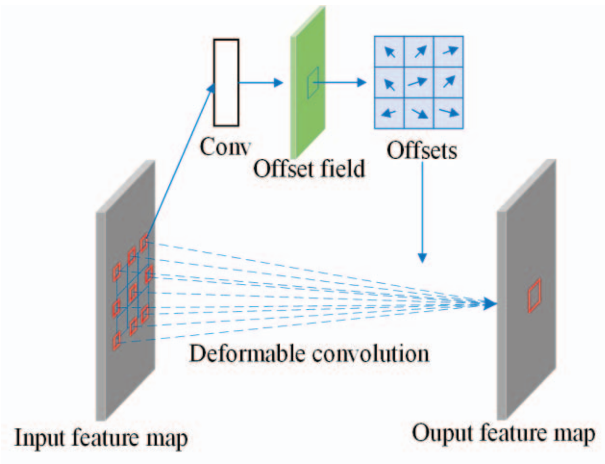

Fig. 3. Deformable convolution process.

B. *Receptive Field Enhancement*

In real industrial environments, variations in ambient light sources, industrial noise and other uncontrollable factors can affect steel surface defect images. This creates a large volume of uncontrollable noise that can adversely affect the detection of certain defects whose characteristics are not obvious, leading to missed inspections. Small cracks, for example, do not have obvious features and are therefore easily masked by noise, resulting in missed detections. In order to effectively suppress noise and enhance robustness in complex scenarios, we design an RFE module added between the backbone and the neck. This design aims to enhance the perceptual range of the feature tensors, thus improving the network's ability to capture long-distance dependencies. The RFE module is convolved at different expansion rates, utilizing the full potential of the perceptual field of the feature tensors. Inspired by TridentNet, the network employs four branches with different expansion rates to capture multi-scale information and different dependency ranges.

All branches have common weights, the only difference is that they have different receptive fields. The RFE module can be divided into two components: multi-branching based on extended convolution and the weighted layer. As illustrated in Fig. 4, the multi-branching component uses dilated convolutions with rates 1, 2 and 3, where all convolutions use a fixed convolution kernel size of 3x3. Additionally, residual connections are incorporated to mitigate the issues of exploding and vanishing gradients during training.

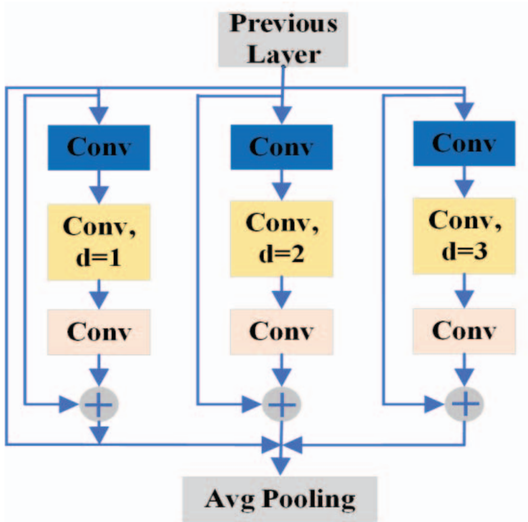

Fig. 4. Structure of the RFE module.

C. *Bi-level Routing Attention*

The outcomes of the experiments demonstrate that the detection accuracy of rolled-in scale and cracks is relatively low. The common characteristics of these two defects are relatively small size and inconspicuous features, which increase the difficulty of detection. Therefore, BRA mechanism is introduced, which uses self-attention mechanism and employs sparse sampling instead of downsampling to tackle the difficulty of identifying small targets in images. The traditional transformer model's numerous parameters result in high computational costs, increased memory usage, and a negative impact on the speed of image detection when directly applied to the entire global range of the feature map. To tackle this issue, researchers have introduced sparse attention in the transformer. In other words, each query concentrates on only a small portion of key-value pairs, rather than the entire set. For example, restricting attentional operations to local windows, axial stripes, or within dilated windows in the feature map, rather than applying them globally, classifies such methods as hand-crafted sparse patterns. Another class of method deals with adaptive sparse patterns. For example, Chen et al. introduced a deformable patch module that adaptively segmented images into patches with varying positions and dimensions in a data-driven manner, deviating from the conventional fixed patch segmentation. To boost the feature extraction capability for small targets, this paper introduces BRA mechanism in the YOLOv5n network model. The general framework of the BRA module is illustrated in Fig. 5, and BRA's main principle is to filter out irrelevant key-value pairs for the broader regions of the feature graph. This is achieved through the construction and selective pruning of region-level directed graphs.

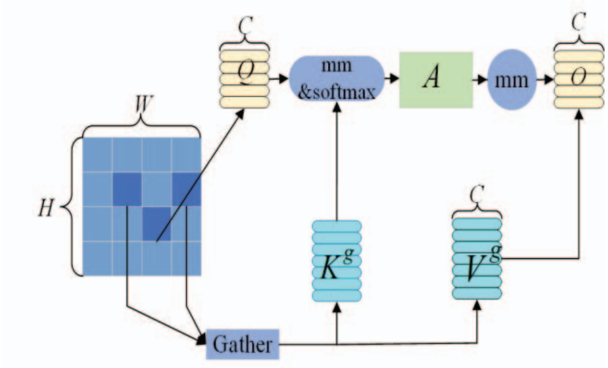

Fig. 5. Structure of the BRA module.

First, the input data representation is segmented into S*S disjoint regions, and This is achieved by reshaping X as X^r. We derive the query, key, and value from X^r with the following formula:

$$Q = X^r W^q, K = X^r W^k, V = X^r W^v \tag{2}$$

In (2), $W^q, W^k, W^v \in R^{C\times C}$ are weights of Q (Query), K (Key), and V (Value), respectively. It is then necessary to identify the portion of each region of interest, i.e., by first calculating the mean Q^r and K^r of Q and K within the region. Subsequently, the inter-regional correlation matrix between them is calculated:

$$A^r = Q^r (K^r)^T \tag{3}$$

Subsequently, we prune the correlation map by keeping only the top-k connections in each region. A matrix of routing indices is obtained, $I^r \in N^{S^2\times K}$,applying the topk operator along the rows:

$$I^r = topkInder(A^r) \tag{4}$$

Hence, the i^{th} row of I^r contains k indices of the most relevant region for the i^{th} region. By utilizing the region-to-region routing index matrix I^r , fine-grained token-to-token attention can then be applied. We then collect the key and value tensor.

$$K^g = gather(K, I^r),\ V^g = gather(V, I^r) \tag{5}$$

Then using the attention operation on the aggregated K^g, V^g yields:

$$\begin{aligned} O &= Attention(Q, K^g, V^g) + LCE(V) \\ &= \text{soft}max\left(\frac{Q(K^g)^T}{\sqrt{C}}\right)V^g + LCE(V) \end{aligned} \tag{6}$$

where $LCE(V)$ represents the local context enhancement term. Consequently, the BRA module can capture not only the image context feature information but also significantly reduce computational complexity.

III. Experiments and Result Analysis

A. Dataset Description

In this paper, Experiments on detecting defects on steel surfaces are conducted using the publicly available NEU-DET dataset. The dataset contains six different defect types: cracks (Cr), inclusions (In), patches (Pa), pitted surfaces (PS), rolled scales (RS) and scratches (Sc). Each defect type contains 300 samples, and thus a total of 1800 images. The defect images of the six categories are randomly assigned in a ratio of 8:1:1. After this division, the training set consists of 1440 samples, the validation set includes 180 samples, and the test set comprises 180 samples.

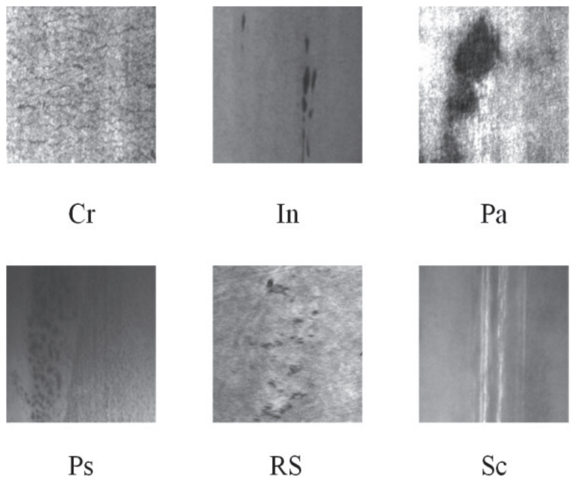

Fig. 6. Sample images of steel surface defects.

As shown in Fig. 6, the overall image of steel defects has a low grey value and thus easily blurs the defect features due to light variations and other industrial noise. The scale and shape of the defects show significant visual differences. In addition, the NEU-DET dataset contains many small defects that make detection more difficult. However, traditional target detection methods are difficult to cope with these challenges, resulting in low detection accuracy.

B. Experimental environment and evaluation Metrics

The experimental operating system is Ubuntu 20.04.6 LTS, and the graphics card is the NVIDIA GeForce RTX 3090 with 128 GB memory. The CUDA version is 11.8. The network model is built using the PyTorch framework, and the PyTorch version is 1.10.0.

TABLE I. ABLATION EXPERIMENT RESULTS

Model	mAP (%)	FPS	AP (%)						Params/M
			Cr	In	Pa	PS	RS	Sc	
Faster-RCNN	79.6	25.6	47.1	84.4	95.1	85.1	**70.3**	95.7	108.4
YOLOv5n	77.0	120.2	42.3	85.3	95.1	84.7	61.8	93.0	**3.9**
YOLOv7-tiny	67.3	**125.6**	33.1	66.2	84.9	**96.9**	52.7	70.0	12.3
YOLOv8n	77.5	90.1	44.9	83.7	94.5	83.9	67.8	90.4	6.3
SDDF-Net *	78.8	-	**61.9**	82.0	**96.7**	78.7	63.3	90.4	10.0
ES-Net[7]*	79.1	53.0	56.0	87.6	88.3	87.4	60.4	94.9	54.2
ETDNet[8]*	80.2	49	-	-	-	-	-	-	6.99
DIN[2]*	80.6	-	61.5	85.9	92.9	90.3	64.8	88.1	-
ours	**82.1**	60.4	53.6	**88.1**	92.6	92.2	68.9	**97.4**	4.6

This experiment evaluates model performance using mean average precision (mAP), frames per second (FPS), and the number of parameters (Params). The mAP represents the average of the mean accuracy for detecting each category. FPS represents how many images are detected per second when using the trained algorithm for prediction. The number of parameters is a measure of the model's size and is usually used to indicate the complexity of the model. A larger number of parameters means that the model has more degrees of freedom to adapt to the data but may also lead to overfitting problems.

C. Ablation study

To assess the effectiveness of the improvements proposed in this paper on the detection algorithm and to determine whether there is any influence between the modules, we perform ablation studies with the dataset and compare the experimental results with YOLOv5n as the comparative model. The outcomes of the experiment are displayed in TABLE II.

TABLE II. ABLATION EXPERIMENT RESULTS

Methods	DCN	RFE	BRA	mAP (%)	FPS
Baseline	×	×	×	77.0	120.4
1	✓	×	×	79.8	90.4
2	×	✓	×	78.6	80.4
3	×	×	✓	79.0	74.4
4	✓	×	✓	80.6	65.1
5	×	✓	✓	79.8	70.0
6	✓	✓	×	80.9	76.2
RDB - YOLO	✓	✓	✓	82.1	60.4

To tackle the challenges in detecting defects on steel surfaces, such as interference from light variations, ambient noise, and the irregular shapes of defects, an enhanced YOLOv5 algorithm is proposed. This algorithm introduces deformable convolution, RFE, and BRA. TABLE II. shows that after the introduction of DCN, the precision of detecting of mAP is significantly improved by 2.8% over the initial YOLOv5n model. The introduction of DCN aims to enhance the model's capability to fit irregular defects. As evident from the TABLE II. , the defect detection accuracies for most defects increases after introducing DCN. This experimental segment confirms the effectiveness of introducing DCN. To mitigate the impacts of light variations and ambient noise, the Receptive Field Enhancement module is used to increase the receptive field, thereby enhancing the capability to capture distant dependencies. The results of the experiments suggest that the inclusion of the RFE module leads to additional improvements in mAP, with increases of 1.6%. To enhance the detection of small target defects, this paper incorporates BRA, the core module of BiFormer based on the transformer variant, into the YOLOv5 network model. The experimental results demonstrate that the model, enhanced by the inclusion of the BRA module, exhibits improvements of 2.0% compared to the previous mAP.

In order to fully evaluate the detection performance of RDB-YOLO, we compare it with four general methods and four well-known methods for detecting surface defects on steel. The experimental results are depicted in TABLE I. , where the experimental outcomes without '*' undergo retraining with us and those with '*' are quoted from academic papers. '-' indicates that no relevant data is included in this paper. As demonstrated by TABLE I. , our RDB-YOLO achieves the highest mAP of 82.1%. YOLOv7-tiny has the fastest FPS detection at 125.6, but the low mAP of 67.3%. The two-stage detector Faster-RCNN achieves a mAP of 79.6 but the FPS is low, and the number of model parameters is large. In comparison to the baseline model YOLOv5, we improve our mAP by 5.1%, although at a reduced FPS and a slightly larger number of parameters. Comparison to the most recent work on the NEU-DET dataset, such as ES-Net、 ETDNet and DIN, the mAP of our RDB-YOLO is still 3.0%,1.9% and 1.5% higher than them and has the smallest number of parameters. Therefore, we can conclude that our model surpasses current leading target detection algorithms, considering both accuracy and parameter count.

D. Additional Experiment

In order to further demonstrate the superiority of the BRA attention mechanism in detecting surface defects on strip steel, we conduct comparative experiments with other common attention mechanisms, and the experimental results are shown in TABLE III. From the data, it is apparent that our BRA-YOLOv5n also achieves the highest 79.0% mAP.

TABLE III. COMPARISON OF COMMON ATTENTION MECHANISM

Method	mAP (%)	AP (%)					
		Cr	In	Pa	PS	RS	Sc
YOLOv5	77.0	42.3	85.3	95.1	84.7	61.8	93.0
SE-YOLOv5n	78.4	61.6	84.1	88.3	81.7	63.5	91.3
CBAM-YOLOv5n	77.3	65.1	80.2	87.0	82.1	56.1	93.1
CA-YOLOv5n	77.4	61.8	83.5	88.2	81.1	58.3	92.7
BRA-YOLOv5n	79.0	57.3	86.8	96.1	77.5	60.8	95.7

IV. CONCLUSION

We propose a detection of surface irregularities in steel approach built on YOLOv5n. It introduces DCN to fit irregular defects, combines the BRA attention mechanism to improve the accuracy of small defects detection, and adds RFE to suppress noise. The RDB-YOLO model is tested on the publicly available NEU-DET dataset. Experimental results indicate that the proposed method significantly enhances detection accuracy, reduces the detection error rate more effectively than other algorithms, and has lighter weight parameters. However, the method proposed in this paper still has some limitations, and the attention mechanism added in network inevitably increases the computational demand. In subsequent studies, we will optimize the algorithm to further improve the detection efficiency.

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